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MILITARY SPECIFICATION

CARRIERS, FULL-TRACKED: M548A1 and M730A1

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of unarmored full-tracked, diesel powered vehicles. One vehicle is designed for cross country and amphibious operation, as a guided missile carrier, the other vehicle is for cross country operation only, as cargo and ammunition support carrier.

1.2 Classification. Vehicles shall be one of the following models (see 6.2).

Model	Army drawing	Vehicle type
M548A1	8750029	Carrier, Cargo, Full-tracked
M730A1	8750067	Carrier, Guided Missile Equipment

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2320

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SPECIFICATIONS
FEDERAL

VV-F-800	- Fuel Oil, Diesel

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.
MIL-C-62015	- Carrier, Cargo, Full-tracked, M548; Processing for Storage and Shipment of.
MIL-C-62053	- Carrier, Guided Missile Equipment, M730 and M730A1 Processing for Storage and Shipment of.
MIL-T-62099	- Transmission, Hydraulic, Modified Automatic TX-100-1.
MIL-E-62140	- Engine Diesel: 6 Cylinder, V-type, 210 Hp.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-1275	- Electrical Circuit, 28 Volt DC Transient Characteristics for Military Vehicles.
MIL-STD-1474	- Noise Limits for Army Materiel.

2.1.2 Other Government documents, and Government drawings and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

8750029	- Carrier, Cargo, M548A1
8750067	- Carrier, Guided Missile Equipment M730A1.

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10942803 - Lubrication Chart, Production, Fuel and Coolant.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article (preproduction). The contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.3). First article samples shall be inspected by the contractor under the surveillance of the Government to determine conformance to the quality assurance provisions of this specification. First article samples shall be fully representative of units to be supplied from production tooling and facilities.

3.2 Initial production vehicles. Vehicles manufactured at initiation of quantity production shall be inspected, and shall pass, initial production inspection (see 4.4). Initial production vehicles shall be representative of vehicles to be furnished under this specification and shall include all current modifications. No modification shall be applied to subsequent vehicles without approval of the Government.

3.2.1 Reliability. Reliability shall be not less than 750 Mean Miles Between Failure (MMBF) during the first 5,000 miles, when operated in accordance with the operational profile (see 3.2.4). For calculating MMBF, the definition in the M113A2 family 50 mile mission failure scoring criteria shall be used (see 6.11).

3.2.2 Durability. The vehicle with rated payload shall have a 50 percent probability of completing the first 5,000 test miles without replacement, rebuild, or major overhaul of the engine, transmission, transfer gear unit, differential steer unit, or final drives, and without repair or rebuild of the hull, when operated in accordance with the operational profile (see 3.2.4).

3.2.3 Maintainability. Total maintenance, excluding driver/crew checks and services, shall not exceed 175 manhours during 5,000 miles of operation in accordance with the operational profile (see 3.2.4). This

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equates to a Maintenance Ratio (MR) of 0.35 at 10 operational miles, equivalent to one operational hour. The scheduled maintenance interval shall be three months or 750 miles, whichever comes first.

3.2.4 Operational profile. Test mileage shall be apportioned as follows:

a. 20 percent hard-surfaced roads
b. 40 percent secondary roads
c. 40 percent cross country

NOTE: When vehicle is to be operated as a prime mover (equipped with air brake kit), mileage during operation on hard-surfaced roads and cross country shall include 200 miles in each mode towing a 14,000 pound GVW wheeled trailer for M548A1 and 3,000 pounds for M730A1.

3.3 Materials. Materials shall be as specified herein and in referenced drawings, specifications, and standards. Materials shall be free of defects and imperfections that might affect the serviceability and function of the finished product (see 6.10).

3.3.1 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications or drawings.

3.3.2 Qualified products. The contractor shall be responsible for using parts and assemblies from Qualified Products Lists (QPL's) whenever available. Contractor's inspection records shall specifically list all QPL items by number and date of the QPL, name of contractor and part or drawing number(s). When parts and assemblies are approved as qualified products, but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document and name of contractor.

3.4 Construction. The vehicle shall be constructed in accordance with the following drawings, as applicable, and applicable specifications and standards (see 3.1). If more than one unit of a particular component is used, the components used shall be identical in make, material and quality. Riveting, welding practices and their quality shall be the same on each vehicle. Completed vehicles shall meet all requirements specified herein, and be of a quality that assures a vehicle free of defects that would compromise, limit, or reduce performance.

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Model	Drawing number
M548A1	8750029
M730A1	8750067

3.4.1 Bearing seals. When fording, operating amphibiously (M730A1 only), or in mud, sand or snow, bearing seals shall prevent the entrance of foreign matter into the bearings, which are exposed to contamination during these operations. Bearing seals shall restrict the leaking of lubricants from bearings.

3.4.2 Special kits. Provisions shall be made for installation of special kits including winterization equipment. When specified (see 6.2), special kits shall be furnished as equipment. Special kits, when installed, shall fit properly without interference or misalinement and meet applicable environmental and performance requirements specified herein.

3.4.2.1 Winterization kit (M730A1). When installed, the winterization kit shall supplement the personnel heater by providing insulation for the vehicle cab, frost shields for the cab door windows, and an insulated cab cover assembly to permit vehicle operation in ambient temperatures from minus 25°F to minus 65°F. The insulated soft cab cover shall resist the entrance of water, dust and snow beyond the extent commensurate with the design characteristics. When installed, the engine coolant heater system shall assure engine starting and operation in ambient air temperature from minus 25°F to minus 65°F. The engine coolant heater shall be capable of operating for a minimum of twelve hours on battery power alone without discharging the battery to a level that would preclude satisfactory starting of the engine.

3.4.2.2 Cab personnel heater kit (primary) (M548A1). When installed, the (primary) cab personnel heater kit shall provide heat for personnel and windshield defrosting in ambient temperatures down to minus 25°F.

3.4.2.3 Cab personnel heater kit (secondary) (M548A1). When installed, the (secondary) cab personnel heater kit shall supplement the (primary) cab personnel heater and insure heat for personnel and windshield defrosting during operations in ambient temperatures from minus 25°F to minus 65°F. The insulated hard top cab cover shall resist the entrance of water, dust and snow beyond the extent commensurate with the design characteristics.

3.4.2.4 Cargo area personnel heater kit (primary) (M548A1). When installed, the (primary) cargo area personnel heater kit shall provide heat for personnel in the rear cargo area in ambient temperatures down to minus 25°F.

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3.4.2.5 Cargo area personnel heater kit (secondary) (M548A1). When installed, the (secondary) cargo area personnel heater kit shall supplement the (primary) cargo area personnel heater kit during operations in ambient temperatures from minus 25°F to minus 65°F. The insulated cargo cover shall provide protection for personnel and material in the cargo area in ambient air temperatures down to minus 65°F. The insulated cargo cover shall be water, dust and snow proof.

3.4.2.6 Engine coolant heater kit (M548A1). When installed, the engine coolant heater kit shall assure engine starting and operation in ambient air temperatures from minus 25°F to minus 65°F. The engine coolant heater shall be capable of operating for a minimum of twelve hours on battery power alone without discharging the battery to a level that would preclude satisfactory starting of the engine.

3.4.2.7 Air brake kit (M548A1). The air brake kit is required on carriers that are to be used as prime movers of vehicles or equipment that have air brakes. Test mileage apportioned to first article vehicles equipped with air brake kits shall include 200 miles with a towed load of approximately 14,000 pounds in the hard-surfaced road mode and in the cross country mode (see 3.2.4).

3.4.2.8 Material handling kit (M548A1). When installed, the material handling kit shall be capable of loading material weighing up to 1500 pounds into the cargo area. This kit shall be capable of being mounted either left of center, right of center, or in a center location of cargo area, without interference.

3.4.2.9 Machine gun kit (M548A1). When mounted on vehicle, the machine gun kit shall be capable of supporting the machine gun when firing, and allow the gun to rotate 360 degrees to the right and left without interference or binding.

3.4.3 Doors, windows and covers. All doors, windows and covers shall conform to applicable drawings. When installed, they shall fit and be water, dust and snow resistant, and function without interference or misalinement.

3.4.4 Engine. The engine shall conform to MIL-E-62140 and shall operate throughout all speed ranges without evidence of cooling system, lubricant, or fuel leakage.

3.4.4.1 Engine air induction system. The air induction system, as installed, shall prevent entrance of foreign matter in conformance with applicable drawings.

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3.4.5 Exhaust systems. The engine exhaust system, as installed, and the personnel heater exhaust system, when installed, shall not permit concentration of toxic fumes in the driver and crew compartments to exceed 50 parts per million (ppm) for carbon monoxide (CO) or 5 ppm for oxides of nitrogen (NO).

3.4.6 Fuel system. The fuel system shall maintain a continuous supply of fuel to the engine when ascending and descending 60 percent grades in forward and reverse gear, and when vehicle is being operated left side or right side up on 30 percent side slopes.

3.4.6.1 Fuel tank. Provision shall be made to assure the internal cleanliness of the fuel tank prior to initial fueling of vehicle. The fuel tank shall be capable of receiving 50 gallons of fuel per minute.

3.4.6.2 Fuel lines. Fuel lines and connections shall be internally clean prior to initial fueling and servicing of vehicle. All fuel lines, connections, and tanks shall free of leaks.

3.4.6.3 Rack linkage. With rack linkage installed in accordance with applicable drawings, and with the accelerator pushed against the stop, the rack shall be at full open position.

3.4.7 Transmission, transfer gearcase, and control differential. The transmission shall conform to MIL-T-62099. When installed in vehicle, the transmission and controlled differential shall operate throughout all gear ranges without evidence of leakage.

3.4.8 Electrical system.

3.4.8.1 Generating system voltage (alternator). After no less than 20 minutes of operation with headlights on, regulator control shall be observed. Alternator shall be operated at 4000 revolutions per minute (rpm) $\pm$ 100 rpm with an output of 50 amperes $\pm$ 5 amperes, and the voltage regulator rheostat adjusted to provide 28 volts direct current (Vdc) at the vehicle batteries. Voltage at the vehicle batteries shall not increase more than 0.5 Vdc when the load is removed.

3.4.8.2 Lights. Internal and external lights specified on applicable drawings shall operate, and all electrical contacts and connections shall maintain positive contact, under all vehicle operating conditions.

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3.4.8.3 Communication and electrical circuits. Communication and electrical circuits shall maintain continuity from end to end, without shorts from pin to pin, or from pin to connector shell.

3.4.8.4 System voltage characteristics. Electrical system voltage characteristics shall fall within the limits for transient and steady state voltage requirements specified in MIL-STD-1275.

3.4.8.5 Electromagnetic interference suppression. The vehicle, when operating, shall meet the requirements for EMI/EMC suppression in accordance with class A3 and C1 of MIL-STD-461.

3.4.8.6 Heaters. The personnel and cargo area heaters shall start and operate satisfactorily on 20 Vdc, using diesel fuel conforming to VV-F-800.

3.4.9 Controls. All electrical, mechanical and hydraulic controls shall operate without malfunction throughout all ranges of operation under all vehicle operating conditions.

3.4.10 Adjustment mechanisms. All adjustment mechanisms shall function properly and maintain adjustment within required settings during all vehicle operating conditions.

3.4.11 Stowed material. All interior and exterior On Vehicle Material (OVM) shall be stowed (combat loaded) on the vehicle, in spaces provided, to assure that items will not interfere with operation of vehicle and components.

3.4.12 Welding repairs. Welding repairs of any kind shall be made only when, and to the extent, specifically authorized by the procuring activity (see 6.2).

3.4.13 Folding blast covers (M730A1). The folding blast covers shall provide protection for the crew compartment when launching a missile. The blast covers shall fold open and remain open to permit vehicle operation without cover closing on operator.

3.4.14 Tailgate (M730A1). The tailgate shall hinge downward and support the weight of two men during manual missile loading of the launcher. When closed the tailgate shall be sealed to restrict the entrance of water into the vehicle during fording and flotation operations.

3.4.15 Air transportability (M730A1). The vehicle when mated with its missile pallet shall be capable of Phase II transport in a C-130 aircraft.

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3.5 Break-in run. Unless otherwise specified, prior to being subjected to testing, each vehicle shall be given a break-in run (see 4.5). Break-in run is not intended to be a part of any test but is carried out by the contractor to insure that the vehicle is in a safe, operable condition prior to testing. A combat-loaded condition is not required. When specified (see 6.2), the break-in run shall be simulated (see 4.5.3).

3.6 Performance. A complete vehicle, loaded with rated payload, serviced as specified in 3.5.1, and equipped for existing climatic conditions, shall operate as specified herein without special equipment. Performance shall be demonstrated on smooth, level, hard-surfaced roads free of loose material, except that a section of road shall be as shown in figure 1.

3.6.1 Environmental conditons. The complete vehicle with rated payload shall operate in ambient air temperature of plus 125°F to minus 25°F and, with special equipment installed, to minus 65°F. The complete vehicle shall withstand storage at minus 80°F without deterioration that may cause failure of any component part of the vehicle.

3.6.2 Cooling systems.

3.6.2.1 Engine cooling. With the vehicle operating in an ambient temperature of 125°F, the engine cooling system shall maintain engine coolant at not more than 230°F, measured at a point in the engine outlet to radiator. Radiator pressure cap shall relieve between 13 and 18 pounds per square inch gage (psig).

3.6.2.2 Transmission, lubricant cooling. With the vehicle operating in an ambient temperature of 125°F, the transmission lubricant cooling system shall maintain lubricant temperature at not more than 325°F, measured out of transmission into cooler, except at torque converter speed ratios less than four to ten.

3.6.2.3 Differential oil cooler. With the vehicle operating in an ambient temperature of 125°F, the differential lubricant cooler shall maintain lubricant temperature below 300°F, measured out of differential into cooler.

3.6.3 Bilge pumps. The bilge pump(s), installed in accordance with applicable drawings, shall remove water from the vehicle at a minimum rate of 35 gallons per minute (gpm) each with a discharge head of 4 1/2 feet, and with pump motor operating on 27.5 Vdc impressed at the motor.

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3.6.4 Speeds and vehicle control. The vehicle shall sustain speeds up to 32 mph, and a maximum speed of 42 mph after 400 miles of operation, without damage to power plant or suspension system. With vehicle traveling between 15 mph and maximum speed, drift shall not exceed 3 feet in 100 feet without corrective effort by the driver.

3.6.5 Acceleration. With the vehicle initially at a standstill, transmission in 1-3 range, the vehicle shall accelerate from 0 to 20 mph in not more than 18 seconds.

3.6.6 Engine starting. With the vehicle parked in the following positions, the engine shall be capable of restarting within 1 minute after being stopped for not less than 1 minute after being operated for not less than 2 minutes under no load between 750 and 1,000 rpm:

a. Headed upgrade on 60% grade.
b. Headed downgrade on 60% grade.
c. Left side up on 30% slope.
d. Right side up on 30% slope.

3.6.7 Braking.

3.6.7.1 Stopping and vehicle control. Vehicle shall stop within a distance of 40 feet from point of service brake application when traveling at a speed of 20 mph on smooth, level, hard-surfaced roads free of loose material. Drift shall not exceed 3 feet in stopping distance. During stopping, the vehicle being under driver's individual control, steering action is permitted.

3.6.7.2 Holding. With brakes locked for parking and with transmission in neutral, the vehicle shall hold stationary when parked on a 60 percent longitudinal grade. For this operation vehicle shall be loaded with rated payload or towing a trailer.

3.6.8 Turning. Vehicle, on smooth, level, hard-surfaced road or track, with differential steer, shall make continuous, full 360 degree turns to right and to left, with rear point of outer edge of vehicle at all times in contact with or within a 50 foot diameter circle. When in pivot steer, the vehicle shall pivot within a 28 foot diameter circle. The applied effort to the steering lever shall not exceed 100 pounds to pivot the vehicle.

3.6.9 Water operation.

3.6.9.1 Shallow water crossing. The vehicle shall ford shallow, hard-bottom salt or fresh water channels up to a depth of 40 inches. Leakage of water into the hull shall not exceed 3 quarts per minute.

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3.6.9.2 Amphibious operation (M730A1). With flotation kit installed, the M730A1 vehicle shall negotiate water of unlimited depth with height of waves not exceeding 1 foot. When afloat in calm water not less than 15 feet deep, a maximum speed of not less than 3 mph shall be attained. The vehicle shall be capable of operating within a 30 foot corridor when crossing calm, open water. Leakage of water into the hull shall not exceed 3 quarts per minute.

3.6.10 Trench crossing. Vehicle shall cross trenches up to 36 inches in depth, and up to 66 inches in width, without damage to vehicle.

3.6.11 Vertical obstacles. Vehicle, operating at low speeds in forward gear, shall cross vertical obstacles (wall) up to 24 inches in height and, in reverse gear, obstacles up to 14 inches in height without damage to vehicle.

3.6.12 Grade and slope operation. Vehicle shall ascend 60 percent grades in forward and reverse gear, from a standstill position on the grade, without stalling or damage to power plant or power train. Vehicle shall operate without malfunction or damage on 30 percent side slopes, with right and then left side of vehicle up-slope. Proper oil pressure and temperature shall be maintained in the engine, transmission, transfer gear unit, and controlled differential during all grade and slope operation. Oil pressure observation shall be made with engine operating at 600 to 800 rpm.

3.6.13 Suspension. The vehicle suspension shall operate over a minimum of three 6x6 inch obstacles, spaced 30 feet apart, at a speed of 18 to 22 mph, without bottoming out of the track drive sprockets, roadarm bump stops, or idler wheels, with either side or both tracks, engaging the obstacles.

3.6.14 Winch. The power driven winch shall be capable of operating at a rated line pull of 20,000 pounds, at a winding speed of not less than 15 feet per minute, using a single line on a bare drum. The winch shall be capable of 4 consecutive self recovery operations. Sufficient time interval between operations for winch cooling is permitted. The winch shall be capable of stopping and holding the load with the power source disconnected.

3.6.14.1 Winch controls. The cab mounted controls shall engage the prop-shaft drive for operation of the winch. The manually operated device furnished shall permit free spooling of the winch drum.

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3.6.14.2 Winch shear safety pin. The installed shear pin shall preclude damage to the winch if the line pull exceeds 24,000 pounds at a winding speed of 1.8 feet per minute.

3.7 Painting, marking and data plates.

3.7.1 Painting. Exterior and interior of vehicle and component assemblies and parts which require painting shall be prepared and painted in accordance with MIL-STD-193, MIL-E-52798 or MIL-E-52835. Exterior color shall be forest green.

3.7.2 Marking. Vehicle marking shall be in accordance with MIL-STD-642. Parts and components shall be marked in accordance with MIL-STD-130.

3.7.3 Name, shipping and service data plates. Plates shall meet the requirements of MIL-P-514, or applicable drawings.

3.8 Noise level. With a sound level meter (see 4.7.32) located 4 inches forward of the engine compartment rear bulkhead, 4 inches to the left of engine compartment left bulkhead, and 12 inches below the inside surface of the top plate, and the vehicle traveling at 30 mph in a 1-3 gear range on a hard-surfaced straight road with all doors, hatches and windows closed, the sound level shall not exceed 119dB(A).

3.8.1 Noise hazard caution sign. The vehicle driver and passenger compartments shall be posted as noise hazardous areas as specified in MIL-STD-1474.

3.9 Fire extinguisher. The fixed fire extinguisher system, when installed in accordance with applicable drawings, shall be readily accessible for servicing and operation internally and externally. Maximum effort for externally discharging system shall not exceed value specified on applicable drawing.

3.10 Workmanship. Workmanship shall be of such quality as to assure that the vehicle and its components are free of defects that compromise, limit or reduce capability in the performance of its intended use. In addition to general appearance, the defects listed in table IV shall be considered to be cause for failure.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsiblity for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified

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in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Materials and qualified products. The contractor's records shall be reviewed to determine conformance to 3.3 and 3.3.2.

4.1.2 Parts and components. Parts, components and assemblies shall be inspected for conformance to requirements of drawings subordinate to those specified in 3.4, and applicable specifications and standards. When applicable, inspection shall also be in accordance with Supplementary Quality Assurance Provisions (SQAP's) or Quality Assurance Requirements (QAR's).

4.2 Classification of inspection. Classification of inspection shall be as follows:

a. First article (preproduction) inspection (see 4.3).
b. Initial production vehicle inspection (see 4.4).
c. Quality conformance inspection (see 4.6).

4.3 First article (preproduction) inspection. First article inspection shall be performed after award of contract and prior to production (see 3.1). Inspection shall be performed on a sample unit which has been produced with equipment and procedures normally used in production. One sample shall be subjected to the examination specified in table IV, all tests specified herein, and examined for conformance to applicable drawings. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.3.1 Failure. Failure to pass any examination or test, or defects in excess of the Acceptable Quality Level (AQL) specified, shall be cause for refusal to grant first article approval.

4.4 Initial production inspection. On beginning production, three vehicles shall undergo, and shall pass, initial production inspection (see 3.2). One vehicle shall undergo first production vehicle inspection, and the second and third vehicles will be subjected to the initial production test.

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4.4.1 First production vehicle inspection.

4.4.1.1 In-process examination. During fabrication of the first production vehicle, an in-process examination will be conducted by representatives of TACOM Product Assurance Directorate to evaluate conformance of materials and workmanship to specified requirements. Examination will be made at the contractor's or subcontractor's facility prior to application of primer and paint. Processing and welding procedures, quality system, and inspection records will be evaluated during this examination.

4.4.1.2 Completed first production vehicle contractor inspection. The first completed production vehicle shall be road tested and inspected by the contractor, as specified in 4.6.5, to determine conformance to contract and specifications. After inspection, the contractor shall submit the acceptable vehicle (and all inspection records and certifications) to the responsible Government inspection element at contractor's plant for preliminary examination.

4.4.1.3 Preliminary examination. The responsible Government inspection element will conduct a preliminary examination, as specified in 4.6.2, of the first completed production vehicle.

4.4.1.4 Provisional inspection. Provisional inspection of the first completed production vehicle will be conducted jointly by representatives of TACOM Product Assurance Directorate and the responsible Government inspection element. Contractor shall provide any required assistance. Testing will be as specified in 4.6.5. Contractor shall make available his inspection plan, records, and certifications pertinent to the vehicle and components.

4.4.1.5 Repair of defects. Defects found as a result of the foregoing inspections shall be corrected by the contractor at no cost to the Government. Failure of the contractor to correct defects promptly shall be cause for suspension of acceptance of vehicles until corrective action has been accomplished and approved by the Government.

4.4.1.6 Vehicle disposition. On completion of first production vehicle inspection, the vehicle shall remain at the manufacturing facility, as a production sample, and shall be the last vehicle shipped on contract. Vehicle may be released sooner at the discretion of the Government. The contractor shall service and maintain vehicle during this period.

4.4.1.7 Final approval and acceptance. Final approval and acceptance of the first production vehicle shall be withheld until second and third vehicles are accepted (see 4.4.2).

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4.4.2 Second and third initial production vehicle inspection. Two additional vehicles (see 4.4) from the first month's production, or the first 20 produced, will be subjected to the initial production test.

4.4.2.1 Initial production test. To determine conformance to Section 3 (inclusive), vehicles will be examined as specified in 4.6.2, and tested as specified in table VI. Subsequently, both vehicles will be tested for 5,000 miles, as specified in table I. Inspection will be performed by the Government, and will require no more than 90 days. The contractor shall expeditiously furnish repair parts, as required to support testing, at no cost to the Government.

TABLE I. 5,000 mile initial production test combat loaded.

	Course	Mileage and speeds
1/	Hard-surfaced roads	1,000 miles at varying speeds up to maximum
	Gravel and dirt roads	2,000 miles at varying speeds up to maximum
	Level cross country	1,000 miles at varying speeds up to maximum
1/	Hilly cross country	1,000 miles at varying speeds up to maximum
	Amphibious operation	One (1) hour minimum at varying speeds up to maximum

1/ Hard-surfaced road and hilly cross country mileage includes 200 miles each with towed load.

4.4.2.2 Reliability verification. To determine conformance to 3.2.1, a point estimate value equal to or greater than 750 miles shall be demonstrated during test. The point estimate value will be computed by dividing the total cumulative test mileage of all the test vehicles by the total cumulative chargeable failures experienced on these vehicles.

4.4.2.3 Durability verification. Conformance to 3.2.2 shall be verified during test at or above a 50 percent confidence level. The durability requirement will be calculated using a binominal approach requiring a minimum test vehicle sample of two vehicles.

4.4.2.4 Maintainability verification. To determine conformance to 3.2.3, a maintenance ratio (MR) equal to or less than that specified in 3.2.3, must be demonstrated during test. The MR will be based upon the total cumulative maintenance manhours with the exceptions of human induced errors and driver/crew checks and services.

4.4.2.5 Test failure. Failure of the vehicle as a result of any defects of a workmanship or materials nature during, or as a result of the 5,000 mile test, shall be cause for rejection of the vehicle. Further, the Government may refuse to continue acceptance of production

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vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the defects. Any defects found during or as a result of the 5,000 mile test, shall be prima facie evidence that all vehicles already accepted prior to completion of the 5,000 mile test are similarly deficient, unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such defects on all vehicles shall be corrected by the contractor at no cost to the Government.

4.5 Break-in run procedure. To determine conformance to 3.5, each vehicle shall be operated on smooth, level, hard-surfaced roads as specified in table II. The undulating portion of the run shall be accomplished twice in each direction at speeds between 5 and 10 mph during division "a," and between 11 and 14.5 mph during division "b." Records shall be kept of vehicle instrument readings.

TABLE II. Speeds and distances for break-in run.

Division of run	Speed - mph	Distance - miles
a	0 to 10	5
b	11 to 20	5
c	21 to 30	10
d	31 to maximum governed	10

4.5.1 Reverse operation. After each division of the break-in run, vehicle shall be stopped, the engine allowed to idle for not less than one minute, and the vehicle driven in reverse for a distance of not less than 50 feet.

4.5.2 Oil pressure and temperature. Prior to the break-in run, the engine shall be operated at 600 to 800 rpm until lubricating oil is at operating pressure and temperature. If at any time during the break-in run the engine has been stopped for at least 15 minutes, the engine shall again be operated, as above, before continuing break-in run. Proper oil pressure and temperature shall be maintained in the engine, transmission, transfer unit, and controlled differential during the break-in run. The differential oil filter element shall be removed, cleaned, and reinstalled following completion of the break-in run.

4.5.3 Simulated break-in run. A simulated break-in run (see table III) may be accomplished on the Army Skid Type Dynamometer in lieu of the run specified in 4.5. The final ten miles of the total break-in run shall be on smooth, level, hard-surfaced roads, except that the vehicle shall be subjected to undulating terrain by being operated over the section of road conforming to figure 1. The undulating portion of the

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test shall be accomplished twice in each direction, at speeds between 5 and 10 mph during division "e" and between 11 and 14.5 mph during division "f." After completion of division "b" and "d," table III, the vehicle shall be operated in reverse as specified in 4.5.1.

TABLE III. Simulated break-in run.

Division of run	Speed (mph)	Miles of test	Test conditions
a	0 to 10	5	Dynamometer
b	11 to 20	5	Dynamometer
c	21 to 30	5	Dynamometer
d	31 to maximum governed	5	Dynamometer
e	21 to 30	5	Track (road)
f	31 to maximum governed	5	Track (road)

4.6 Quality conformance inspection.

4.6.1 Inspection provisions.

4.6.1.1 Lot size. An inspection lot shall consist of all vehicles of one model, produced during an identifiable production period, and submitted at one time for acceptance inspection. Production period shall not exceed one week.

4.6.1.2 Sampling for inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105. Before sampling, the contractor shall 100 percent inspect the first 20 vehicles to establish a process average (see 6.6), to allow normal sampling in accordance with MIL-STD-105.

4.6.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examinations shall be performed against the classification of defects and with the acceptable quality levels (AQL) specified in table IV. The following constitutes a part of the classification of defects (see 6.7).

a. Any weep constitutes a minor defect.
b. Any seep constitutes a minor defect.
c. Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
d. Any droplet that occurs at a static fit, metal to metal or gasket combination shall constitute a major defect.

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TABLE IV. Classification of defects.

Category	Defects	Method of inspection
Major:	AQL 25 defects/100 units	
101	Engine; malfunction, leaks, improper mounting	Visual and functional
102	Transmission; malfunction, leaks, improper mounting	Visual and functional
103	Transfer gear, differential steer, and final drive; malfunction, leaks	Visual and functional
104	Fuel system components; malfunction, leaks 1/	Visual and functional
105	Cooling system components; malfunction, leaks, damage	Visual and functional
106	Instrumentation switches, warning and indicating safety devices; malfunction, damage	Visual and functional
107	Electrical system components and circuits, communication system; malfunction 2/	Visual and functional
108	Suspension system components and adjustments; damage, leaks, improper clearance, misalinement	Visual and functional
109	Tracks; damage, pad separation	Visual and functional
110	Doors, hatches, tailgate covers and windows; improper fit, malfunction, damage	Visual and functional
111	Vision devices and receptacles, damage, malfunction	Visual and functional
112	Bilge pumps; malfunction, damage	Functional
113	Covers, grilles, ducts, vents, panels, stowage boxes, seats, fenders, hardware; malfunction, locking devices, damage	Visual and functional
114	Controls; malfunction, clearance improper	Visual and functional
115	Track drive sprockets and drive shafts; damage, clearance improper	Visual and SIE 3/
116	Adjustment mechanisms; malfunction, clearance improper	Visual and functional
117	Hull; structural and welding defects	Visual and functional
118	Towing pintle, lifting eyes, hooks and brackets; malfunction, defects	Visual and functional
119	Seats, crash pads, and straps; damaged, seat control malfunction	Visual and functional

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TABLE IV. Classification of defects. - Continued

Category	Defects	Method of inspection
120	Fire extinguisher system control; seals missing, components damage	Visual
121	Wiring, tubing or hoses; defective, damage	Visual
Minor:	AQL 150 defects/100 units	
201	Engine; improper component assembly	Visual and functional
202	Transmission; improper component assembly or installation	Visual and functional
203	Transfer gear, differential steer and final drive units; improper component assembly or installation	Visual and functional
204	Fuel system components; improper line clearance, assembly or installation 1/	Visual and functional
205	Cooling system components; improper assembly or installation	Visual and functional
206	Instrumentation, switches, warning, indicating and safety devices; improper assembly or installation	Visual and functional
207	Electrical, system components and circuits, communication system; improper assembly or installation	Visual and functional
208	Suspension, system components and adjustments; improper installation or assembly, bolt or nut torque	Visual and functional SIE 3/
209	Tracks; tension, installation or assembly improper, including bolt torque, damage	Visual and SIE 3/
210	Doors, hatches, covers and windows; improper assembly or installation	Visual and functional
211	Vision devices and receptacles; improper assembly, installation or clearance	Visual and functional
212	Bilge pumps; improper assembly, installation	Functional
213	Covers, grilles, ducts, vents, panels, stowage boxes, seats, fenders, hardware; improper fits, adjustment, installation or defective welds, seals, adhesion, hardware or sheet metal	Visual and functional
214	Controls; improper assembly, or installation	Visual and SIE 3/
215	Track drive sprockets and drive shafts; improper assembly, installation or bolt torque	Visual and SIE 3/

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TABLE IV. Classification of defects. - Continued

Category	Defects	Method of inspection
216	Adjustment mechanisms; improper assembly, or installation	Visual and functional
217	Towing pintle, lifting eyes, hooks and brackets; improper fit, installation assembly, or adhesion welding defects, damaged	Visual and functional
218	Seats, crash pads, and straps; data card incomplete, improper assembly or installation	Visual
219	Fire extinguisher system control; data card incomplete, improper assembly, installation, coding or protection	Visual
220	Wiring, tubing or hoses; defective, missing, improper installation, coding or protection	Visual
221	Paint; application or color improper	Visual
222	Tube fittings; defective, missing, improperly installed	Visual
223	Decals, marking data and instruction plates; missing, improper location or size, defective	Visual
224	Lubrication and lubricants; improper application, types and levels	Visual
225	Protective anti-skid coatings; application, materials and coverage improper	Visual
226	Record forms, and publications; missing, improperly processed or secured.	Visual

1/ Fuel leaks constitute a special defect and will result in total inspection of the lot for this defect.
2/ Filament failures are not to be classified as major or minor defects.
3/ SIE = Standard Inspection Equipment.

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4.6.2.1 Unclassified defects. Defects considered to be departures from good workmanship but having no bearing on function, safety, interchangeability, or life shall be noted in writing. Such defects recurring in five consecutive lots, or ten lots or more within a thirty day period, will be added to the minor classification of defects with no increase in AQL.

4.6.2.2 Examination failure. If a vehicle fails to pass any examination specified herein, the Government will stop acceptance until evidence has been provided by the contractor that corrective action has been taken.

4.6.3 Classification of tests. Classification of tests shall be as follows:

a. Acceptance tests (see 4.6.4).
b. Control tests (see 4.6.5).
c. Comparison tests (see 4.6.6).

4.6.4 Acceptance tests. To determine conformance to section 3 (inclusive), after the break-in run (see 4.5), each vehicle shall be examined as specified in 4.6.2, operated a distance of 8 miles or more by the contractor, and subjected to acceptance tests specified in table VI. Except for equipment and component installation, and servicing performed by the contractor as a final processing of vehicles for acceptance, vehicles shall be completely assembled and serviced with fuel, lubricants, and fluids as specified. Vehicles need not be loaded with rated payload, except for holding test and amphibious operation as specified in 4.6.23 and 4.7.26. After testing, the vehicle shall be examined for evidence of fuel or lubricant leakage, or other defects. Such items as floor plates, personnel seat cushions, trim vane, OVE and decals may be omitted from the vehicle during testing.

4.6.4.1 Test failure. If a vehicle fails to pass any acceptance test specified herein, the Government will withhold acceptance of subsequent vehicles until evidence has been provided by the contractor that corrective action has been taken.

4.6.5 Control tests.

4.6.5.1 Frequency. One of the first 10 vehicles produced, and thereafter one per month, shall be selected for control testing after break-in run.

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4.6.5.2 Fifty mile test. Vehicle shall be loaded to full combat weight and operated for a distance of 50 miles over relatively smooth, hard-surfaced roads. During and subsequent to the test, as appropriate, vehicle shall meet the requirements specified for control tests in table VI.

4.6.5.3 Failure. If a vehicle fails to pass any control test specified herein, the Government will withhold acceptance examination and testing on subsequent vehicles until evidence has been provided by the contractor that corrective action has been taken.

4.6.6 Comparison tests. The Government may select vehicles at any time during the contract production period and subject them to Government proving ground testing.

4.6.6.1 Sampling and lot size. Vehicles will be selected on a spot check basis, in a quantity and at a time, specified by the Government.

4.6.6.2 Inspection routine. The sample vehicle will be subjected to the comparison tests specified in table VI, examined as specified in table IV and subjected to the 2000 mile road test specified in table V. Samples will not include any vehicles previously tested.

4.6.6.3 Test failure. Failure of any vehicle to comply with any of the requirements specified herein, or any major defect of a workmanship or materials nature occuring during, or as a result of the tests, may be cause for refusal to continue acceptance of vehicles by the Government until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the condition which caused the rejection.

TABLE V. 2000 mile comparison test, combat loaded.

Course	Mileage and speed
1/ Hard-surfaced or gravel	200 miles at varying speeds up to maximum
Gravel and dirt road	500 miles at varying speeds up to maximum
Level cross country	650 miles at varying speeds up to maximum
1/ Hilly cross country	650 miles at varying speeds up to maximum
Amphibious	One (1) hour minimum at varying speeds up to maximum

1/ Hard-surfaced road and hilly cross country mileage includes 100 miles each with towed load.

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TABLE VI. Classification and location of inspections and tests.

Title	Requirement	First article initial vehicle production 1/	Acceptance 2/	Control 2/	Comparison 1/
Reliability	3.2.1	4.4.2.2			
Durability	3.2.2	4.4.2.3			
Maintainability	3.2.3	4.4.2.4			
Bearing seals	3.4.1	4.7.1	4.7.1		
Special kits tests	3.4.2	4.7.2			
Winterization kit (M730A1)	3.4.2.1	4.7.2.1			4.7.2.1
Cab personnel heater kit (primary) (M548A1)	3.4.2.2	4.7.2.2			
Cab personnel heater kit (secondary) (M548A1)	3.4.2.3	4.7.2.3			
Cargo area personnel heater kit (primary) (M548A1)	3.4.2.4	4.7.2.4			
Cargo area personnel heater kit (secondary) (M548A1)	3.4.2.5	4.7.2.5			
Engine coolant heater kit (M548A1)	3.4.2.6	4.7.2.6			
Air brake kit (M548A1)	3.4.2.7	4.7.2.7			4.7.2.7
Material handling kit (M548A1)	3.4.2.8	4.7.2.8			
Machine gun kit (M548A1)	3.4.2.9	4.7.2.9	4.7.2.9		
Doors, windows, and covers	3.4.3	4.7.3	4.7.3	4.7.3	
Engine	3.4.4	4.7.4	4.7.4		
Engine air induction system	3.4.4.1	4.7.4.1	4.7.4.1		
Exhaust systems	3.4.5	4.7.5		4.7.5	4.7.5
Fuel system, grades and slopes	3.4.6	4.7.6	4.7.6		4.7.6
Fuel tank	3.4.6.1	4.7.6.1	4.7.6.1	4.7.6.1	
Fuel lines	3.4.6.2	4.7.6.2	4.7.6.2		
Rack linkage	3.4.6.3	4.7.6.3	4.7.6.3		
Transmission and control differential	3.4.7	4.7.7	4.7.7		
Generating system voltage	3.4.8.1	4.7.8.1	4.7.8.1		
Lights	3.4.8.2	4.7.8.2	4.7.8.2		
Communication and electrical circuits	3.4.8.3	4.7.8.3	4.7.8.3	4.7.8.3	
System voltage characteristics	3.4.8.4	4.7.8.4			
Electromagnetic interference suppression	3.4.8.5	4.7.8.5			4.7.8.5
Heaters	3.4.8.6	4.7.8.6		4.7.8.6	
Controls	3.4.9	4.7.9	4.7.9		
Adjustment mechanisms	3.4.10	4.7.10	4.7.10		
Stowed material	3.4.11	4.7.11		4.7.11	
Welding repairs	3.4.12	4.7.12	4.7.12		

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TABLE VI. Classification and location of inspections and tests. - Continued

Title	Requirement	First article initial vehicle production 1/	Acceptance 2/	Control 2/	Comparison 1/
Folding blast covers (M730A1)	3.4.13	4.7.13			
Tailgate (M730A1)	3.4.14	4.7.14			
Air transportability	3.4.15	4.7.15			
Break-in run	3.5	4.5			
Environmental conditions	3.6.1	4.7.16			
Engine cooling	3.6.2.1	4.7.17			
Transmission lubricant cooling	3.6.2.2	4.7.17			
Differential oil cooler	3.6.2.3	4.7.17			
Bilge pumps	3.6.3	4.7.18		4.7.18	4.7.18
Speeds and vehicle control	3.6.4	4.7.19	4.7.19		
Acceleration	3.6.5	4.7.20	4.7.20		
Engine starting	3.6.6	4.7.21	4.7.21		4.7.21
Stopping and vehicle control	3.6.7.1	4.7.22	4.7.22		4.7.22
Holding	3.6.7.2	4.7.23	4.7.23 3/	4.7.23	
Turning	3.6.8	4.7.24	4.7.24	4.7.24	
Shallow water crossing	3.6.9.1	4.7.25	4.7.25		
Amphibious operation (M730A1)	3.6.9.2	4.7.26		4.7.26	4.7.26
Trench crossing	3.6.10	4.7.27			4.7.27
Vertical obstacle	3.6.11	4.7.28			4.7.28
Grade and slope operation	3.6.12	4.7.29			4.7.29
Suspension	3.6.13	4.7.30			
Winch	3.6.14	4.7.31			
Noise level	3.8	4.7.32		4.7.32	
Fire extinguisher	3.9	4.7.33		4.7.33	
Workmanship	3.10	4.7.34	4.7.34		
Packaging	Section 5	4.7.35			

1/ Government proving ground.
2/ Manufacturer/contractor facility.
3/ To be performed with the vehicle unloaded.

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4.7 Conformance verification.

4.7.1 Bearing seals test. To determine conformance to 3.4.1, the vehicle shall be operated in the environments and conditions specified. Bearings and seals shall subsequently be examined for lubricant leakage and lubricant contamination.

4.7.2 Special kits examination. To determine conformance to 3.4.2, when special kits are furnished and installed, the kits shall be tested in accordance with 4.7.2. through 4.7.2.9, as applicable.

4.7.2.1 Winterization kit (M730A1). To determine conformance to 3.4.2.1, provisions for installation of winterization kit shall be examined for completeness. When specified, winterization kit shall be installed, the vehicle operated at the specified temperatures, and equipment evaluated for functional requirements.

4.7.2.2 Cab personnel heater kit (primary) (M548A1) test. To determine conformance to 3.4.2.2, the (primary) cab personnel heater kit shall be operated as in use at ambient temperatures to minus 25°F.

4.7.2.3 Cab personnel heater kit (secondary) (M548A1) test. To determine conformance to 3.4.2.3, the (secondary) cab personnel heater kit shall be operated as in use at ambient temperatures from minus 25°F to minus 65°F.

4.7.2.4 Cargo area personnel heater kit (primary) (M548A1) test. To determine conformance to 3.4.2.4, the (primary) cargo area personnel heater kit shall be operated as in use at ambient temperatures to minus 25°F.

4.7.2.5 Cargo area personnel heater kit (secondary) (M548A1) test. To determine conformance to 3.4.2.5, the (secondary) cargo area personnel heater kit shall be operated as in use for the specified performance at ambient temperatures ranging from minus 25°F to minus 65°F.

4.7.2.6 Engine coolant heater kit (M548A1) test. To determine conformance to 3.4.2.6, the engine coolant heater kit shall be operated as in use in ambient air temperatures between minus 25°F to minus 65°F. The engine coolant heater shall be operated on battery power alone for a minimum of twelve hours. The batteries shall then retain sufficient charge to start the engine.

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4.7.2.7 Air brake kit test (M548A1). To determine conformance to 3.4.2.7, provisions for installation of the air brake kit shall be evaluated. When installed, performance shall be determined during road test (see 3.2.4 and table II).

4.7.2.8 Material handling kit test (M548A1). To determine conformance to 3.4.2.8, the material handling kit shall load and unload material as specified. Flexibility of mounting arrangements shall also be confirmed.

4.7.2.9 Machine gun kit test (M548A1). To determine conformance to 3.4.2.9, the machine gun shall be rotated 360 degrees to the right and left and subsequently test fired.

4.7.3 Doors, windows and covers test. To determine conformance to 3.4.3 all doors, windows and covers (including those of special kits when installed) shall be operated and examined for functional requirements. They shall be tested to assure resistance to water, dust and snow.

4.7.4 Engine test. To determine conformance to 3.4.4, the engine shall be operated throughout all speed ranges, and examined for proper installation and absence of leakage of lubricants, fuel and coolants. Prior to assembly and installation, all coolant, lubricant, and fuel lines shall be examined to insure internal cleanliness.

4.7.4.1 Engine air induction systems examination. To determine conformance to 3.4.4.1, the engine air induction system shall be examined for proper installation and absence of leaks.

4.7.5 Exhaust systems toxic fume test. To determine conformance to 3.4.5, the exhaust system shall be examined for proper installation and absence of leaks. Toxic fume levels measured in the driver and crew compartment, shall not exceed specified values. For the test, the vehicle shall be sealed, the engine operated at 900 to 1,000 rpm and the personnel heater, if installed, operated at high heat, for a period of not less than 30 minutes.

4.7.6 Fuel system grades and slopes test. With the vehicle operating on specified grades and slopes, the vehicle engine and fuel system shall be observed to determine conformance to 3.4.6.

4.7.6.1 Fuel tank test. To determine conformance to 3.4.6.1, the fuel tank shall be examined for cleanliness. The fuel tank(s) filled to capacity, and all fuel lines and connections, shall be examined for leakage before and after the testing specified in 4.6.4.

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4.7.6.2 Fuel lines examination. To determine conformance to 3.4.6.2, the fuel lines shall be examined for leaks at all connections and sealed items before and after a road test on smooth, level, hard-surfaced road.

4.7.6.3 Rack linkage examination. To determine conformance to 3.4.6.3, the accelerator pedal shall be operated to full throttle, and the rack examined to assure that full open position has been reached when pedal is in contact with stop.

4.7.7 Transmission, transfer gearcase and control differential test. To determine conformance to 3.4.7, transmission and control differential shall be operated throughout all gear ranges, and examined for proper installation and absence of leaks.

4.7.8 Electrical systems test.

4.7.8.1 Generating system voltage (alternator) test. To determine conformance to 3.4.8.1, with the alternator operating as specified, a voltage measurement shall be made with a voltmeter after described load is removed, and immediately after the road test.

4.7.8.2 Lighting test. All lights shall be operated during and after break-in run throughout all stages of operation, and observed to determine conformance to 3.4.8.2.

4.7.8.3 Communication and electrical circuits test. To determine conformance to 3.4.8.3, electrical circuits supplying current to vehicle equipment shall be tested at all points to assure continuity throughout the circuits. Previously tested circuits shall be functionally tested after installation.

4.7.8.4 System voltage characteristics test. System voltage characteristics shall be evaluated as specified in MIL-STD-1275 to determine conformance to 3.4.8.4.

4.7.8.5 Electromagnetic interference suppression test. To determine conformance to 3.4.8.5, vehicle shall be tested as specified in MIL-STD-462.

4.7.8.6 Heaters voltage test. To determine conformance to 3.4.8.6, verify that the personnel and cargo area heaters start and operate satisfactorily when 20 Vdc is applied at the appropriate connectors.

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4.7.9 Controls test. All controls shall be operated as specified and observed to determine conformance to 3.4.9.

4.7.10 Adjustment mechanisms test. To determine conformance to 3.4.10, all adjustment mechanisms shall be observed during operation, examined for proper adjustment, and adjusted if required.

4.7.11 Stowed material test. To determine conformance to 3.4.11, all manufacturer and depot installed OVM shall be stowed (combat loaded) on the vehicle. All interior and exterior items shall fit in the spaces provided. The material shall be removed after the test. The OVM shall be of the latest configuration available to the contractor. The contractor shall be permitted to unpackage and repackage equipment used for test (for domestic shipment only).

4.7.12 Welding repairs and burning examination. To determine conformance to 3.4.12, all welding repairs and burning operations shall be examined to determine that slag has been removed and approved procedures have been followed.

4.7.13 Folding blast covers (M730A1). To determine conformance to 3.4.13, folding blast covers shall be installed and checked for proper fit and function.

4.7.14 Tailgate (M730A1). To determine conformance to 3.4.14, the hinged tailgate shall be checked for proper installation fit and function.

4.7.15 Air transportability (M730A1). To determine conformance to 3.4.15, examine the vehicle with the missile pallet mounted to assure the specified transportability.

4.7.16 Environmental operation test. To determine conformance to 3.6.1, vehicle shall be stored and operated at the temperatures specified.

4.7.17 Cooling system test. To determine conformance to 3.6.2.1, 3.6.2.2, and 3.6.2.3, the vehicle shall be operated at specified temperatures and cooling system temperatures measured during and after the test. Radiator pressure cap shall be pressure tested and shall relieve pressure within specified range.

4.7.18 Bilge pumps test. To determine conformance to 3.6.3, the vehicle shall have sufficient water in hull to measure bilge pumps operation. Rate of water removal, motor voltage and discharge head shall be measured.

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4.7.19 Speeds and vehicle control test. To determine conformance to 3.6.4, vehicle shall be driven at speeds from 2 to 3 mph for not less than 5 minutes and at specified speeds thereafter, and performance observed.

4.7.20 Acceleration test. To determine conformance to 3.6.5, the vehicle shall be operated as specified and acceleration measured.

4.7.21 Engine starting. To determine conformance to 3.6.6, the vehicle shall be driven up a 60 percent grade, engine stopped for not less than specified time and restarted. Vehicle shall then be operated on 30 percent side slopes and engine stopped for not less than specified time, then restarted.

4.7.22 Stopping and vehicle control test. To determine conformance to 3.6.7.1, the vehicle shall be driven at specified speed. Three stops shall be made and the results shall be averaged to determine acceptance.

4.7.23 Holding test. To determine conformance to 3.6.7.2, the vehicle shall be driven up the specified grade, brakes applied and vehicle observed for movement. Then, with brakes locked for parking, the vehicle shall again be observed for movement. During this test, vehicle shall be combat loaded, or with a simulated load of equal weight placed in a position that will not restrict the flow of engine air intake (or exhaust).

4.7.24 Turning test. To determine conformance to 3.6.8, the vehicle shall be operated as specified, turned to the right and left in full 360 degree turns and performance observed.

4.7.25 Shallow water fording and flotation test. To determine conformance to 3.6.9.1, the vehicle shall be driven into water of specified depth without special equipment. Water accumulation shall not exceed specified amount after 15 minutes of test.

4.7.26 Amphibious operation test (M730A1). To determine conformance to 3.4.1 and 3.6.9.2, the vehicle shall be loaded with rated payload, operated amphibiously and performance observed. The amount of water in the hull and leakage of lubricants from the bearings shall be measured.

4.7.27 Trench crossing test. To determine conformance to 3.6.10, the vehicle shall be operated over trenches of specified depth and width.

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4.7.28 Vertical obstacle test. To determine conformance to 3.6.11, the vehicle shall be operated over vertical obstacles of specified height.

4.7.29 Grade and slope operation. To determine conformance to 3.6.12, the vehicle shall be operated on specified grade and slope.

4.7.30 Suspension test. To determine conformance to 3.6.13, vehicle, operating at a speed of 18 to 22 mph, shall cross specified obstacles without bottoming out.

4.7.31 Winch operation test. To determine conformance to 3.6.14 through 3.6.14.2, the winch shall be operated as specified and performance observed.

4.7.32 Noise level test. To determine conformance to 3.8, the vehicle noise level measured at the location indicated shall not be more than the specified dB, when measured with a sound level meter which meets the requirements of MIL-STD-1474. Driver and passenger compartments shall be examined for signs indicating noise hazard, as specified in 3.8.1.

4.7.33 Fire extinguisher test. To determine conformance to 3.9, the fire extinguisher system shall be examined and functionally tested (see 6.4).

4.7.34 Workmanship. Vehicle shall be examined to determine conformance to 3.10.

4.7.35 Preservation, packaging, and vehicle processing inspection. Material and equipment shall be inspected, prior to shipment, to determine conformance to Section 5 of this specification. Sampling shall be as specified in applicable packaging specification referenced therein.

5. PACKAGING

5.1 Preparation for delivery. Preparation for delivery and storage shall be in accordance with MIL-C-62015 for model M548A1 and MIL-C-62053 for model M730A1.

6. NOTES

6.1 Intended use.

6.1.1 M548A1 carrier. The M548A1 is used as a cargo and ammunition support carrier.

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6.1.2 M730A1 carrier. The M730A1 is used as a guided missile equipment carrier.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Model of carrier required (see 1.2).
c. First article (preproduction model), if required (see 3.1).
d. If winterization equipment is required (see 3.4.2.1).
e. If special kits for M548A1 are required (see 3.4.2.2).
f. If welding repairs are authorized, and extent allowed (see 3.4.12).
g. If simulated break-in run is authorized (see 4.5.3).

6.3 First article. First article samples shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for examination, tests and approval of the first article (see 3.1).

6.4 Safety precautions. Caution should be used in handling carbon dioxide extinguisher cylinders. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher control system since physical injury is highly probable.

6.5 Inspection equipment calibration and inspection. In inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gaging, measuring and test equipment of the required accuracy and precision. Except as otherwise specified, the contractor shall have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

6.6 Process average. Sampling may be initiated if the process average value for the first twenty vehicles inspected is lees than the AQL specified in the classification of defects for major and minor defects.

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of vehicles inspected}} \times 100$$

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If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified AQL.

6.7 Leakage definitions. The following definitions shall be used with the classification of defects for leaks:

a. Weep: Any evidence of fluid beyond the seal.
b. Seep: Any evidence of fluid beyond the seal that does not result in formation of a droplet.
c. Droplet: Any evidence of fluid beyond the seal that results in the formation of a droplet.
d. Drip: Any evidence of fluid beyond the seal where the droplet forms and falls.

6.8 Deficiency sheet. The Government inspector shall verify that a thorough inspection of each vehicle is performed by the contractor for the listed characteristics, and for any departures from good workmanship. The Government inspector shall assure that all defects encountered during inspection are enumerated on deficiency sheet for the vehicle. Defects noted on a deficiency sheet shall contain sufficient description to enable the Government inspector and the contractor's representative to classify the defects in accordance with the classification of defects (see table IV) and definitions contained in MIL-STD-105. Corrective action shall be taken for nonrecurring defects.

6.8.1 Recurring major defects. A major defect is recurring when the same defect occurs more than once in the same sample, or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major defects shall be cause for the entire lot or lots to be inspected for the recurring defects.

6.8.2 Recurring minor defects. A minor defect is recurring if it occurs more than once in the same sample or when the defect occurs in four successive samples. Recurring minor defects shall be cause for the entire lot or lots to be inspected for the recurring defects.

6.9 International standardization agreement. Certain provisions (3.6.9) of this specification are the subject of international standardication agreements NATO STANAG 2805-A (Edition 3), and ABCA-ARMY-STD-151. When amendment, revision or cancellation of this

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specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

6.10 Recycled materials. The use of recycled materials, which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item, shall be encouraged (see 3.3).

6.11 Mission failure scoring criteria. M113A2 Family Vehicles Failure Definition and Scoring Criteria may be obtained from the US Army Tank-Automotive Command, Warren, MI 48090, ATTN: DRSTA-QRA.

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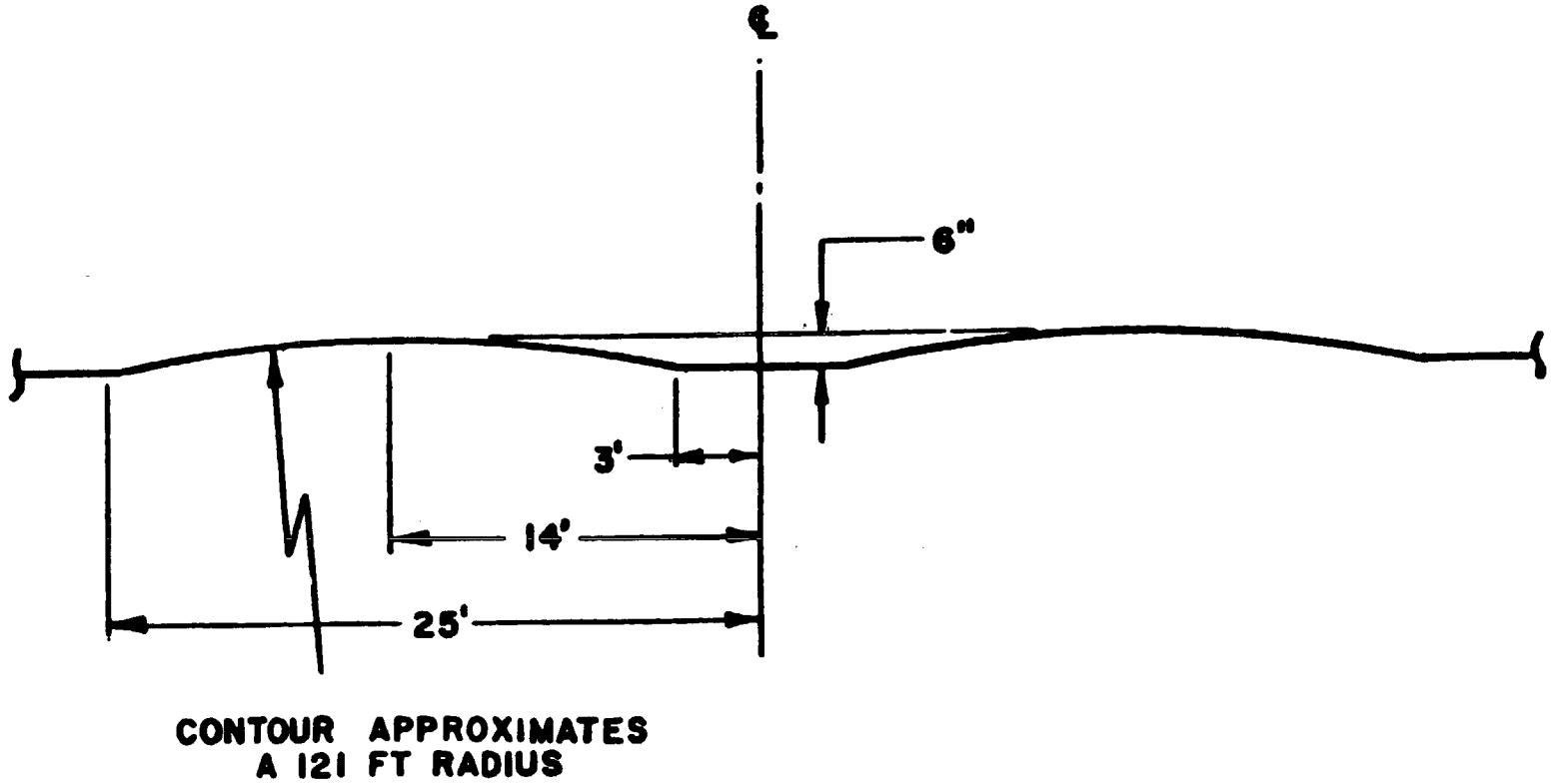


FIGURE 1. Profile of test track (artificial bump).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-62013C(AT); Carriers, Full-Tracked; M548A1 and M730A1

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